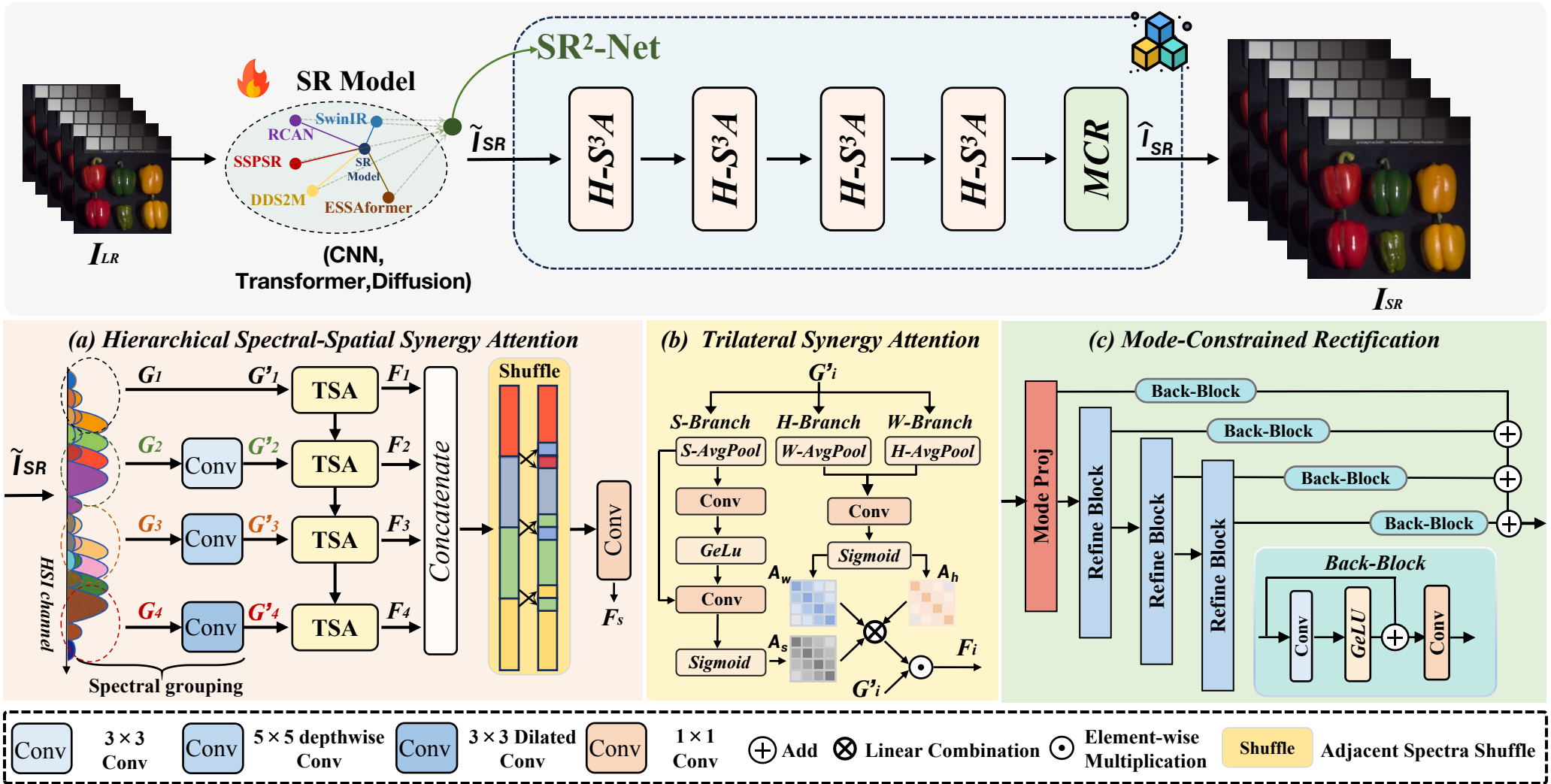

SR Model
SwinIR
RCAN
SSPSR
SR Model
DDS2M
ESSAformer
(CNN, Transformer,Diffusion)
SR²-Net
H-S³A
H-S³A
H-S³A
H-S³A
MCR
(a) Hierarchical Spectral-Spatial Synergy Attention
TSA
Conv
Concatenate
Shuffle
HSI channel
Spectral grouping
(b) Trilateral Synergy Attention
S-Branch
H-Branch
W-Branch
S-AvgPool
W-AvgPool
H-AvgPool
GeLu
Sigmoid
(c) Mode-Constrained Rectification
Mode Proj
Refine Block
Back-Block
GeLU
3 × 3 Conv
5 × 5 depthwise Conv
3 × 3 Dilated Conv
1 × 1 Conv
Add
Linear Combination
Element-wise Multiplication
Adjacent Spectra Shuffle